

Proof of Concept Report
Teeth Bleaching and Hollywood Smile

Waleed Alzamil

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Abstract

This report presents a proof of concept (POC) focusing on teeth bleaching and achieving a Hollywood smile. The project incorporates advanced modules such as segmentation, deep learning approaches, and traditional techniques to deliver aesthetically enhanced dental images. While the teeth bleaching approach yielded excellent results, the Hollywood smile component faced challenges due to the lack of suitable data for deep learning, necessitating the use of a projection-based method. However, this approach was limited by segmentation that treated the entire mouth as a single region rather than segmenting individual teeth, resulting in unacceptable outcomes.

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Chapter 1

Introduction

1.1 Background

Teeth bleaching and the Hollywood smile are popular cosmetic dentistry procedures aimed at enhancing the aesthetic appeal of teeth.

Teeth Bleaching: This process involves whitening teeth by removing stains and discoloration, typically using chemical agents like hydrogen peroxide. It requires precision to ensure uniform results while preserving the natural texture of teeth.

Hollywood Smile: A comprehensive approach that focuses on achieving a perfect smile by reshaping and realigning teeth, often involving veneers, crowns, or advanced imaging techniques. This procedure demands a high level of personalization and aesthetic planning.

In recent years, advanced image processing and deep learning have emerged as transformative tools in achieving these objectives, enabling automated, high-precision segmentation and enhancement of dental images.

1.2 Objectives

The primary goal of this proof of concept (POC) is to explore the potential of deep learning and segmentation techniques in automating and enhancing teeth bleaching and Hollywood smile procedures. Specifically, the objectives include:

- Develop and implement a robust segmentation model to isolate regions of interest in dental images.
- Evaluate the effectiveness of deep learning in achieving uniform teeth bleaching results.

- Explore alternative techniques for Hollywood smile enhancement in the absence of granular segmentation.
- Identify challenges and propose future directions for improvement in dental image processing.

Chapter 2

Methodology

2.1 Segmentation Module

The segmentation process is a critical component of this proof of concept. It leverages a deep learning model to isolate regions of interest in dental images. This module forms the foundation for subsequent processes such as teeth bleaching and Hollywood smile enhancement.

2.1.1 BiSeNet for Segmentation

The BiSeNet -more details in appendixA- model was selected for its ability to balance efficiency and accuracy. It extracts semantic and spatial features from dental images, providing robust segmentation outputs. While it performed well for bleaching, the model's inability to segment individual teeth hindered its application for Hollywood smile enhancements.

2.1.2 Workflow

1. Input images were preprocessed to standardize dimensions and normalize pixel values.
2. The BiSeNet model processed the images, generating segmentation maps of the oral cavity.
3. Outputs were evaluated to ensure sufficient quality for downstream applications.

2.1.3 Challenges

- Lack of granularity in segmentation outputs restricted their use for precise enhancements.
- The absence of individual tooth isolation required alternative approaches for the Hollywood smile process.

2.2 Deep Learning Approaches

Deep learning was utilized exclusively for the segmentation step in the teeth bleaching process. The BiSeNet model provided accurate segmentation masks, isolating the teeth region from the rest of the oral cavity. These masks were then processed using a traditional whitening function, which adjusted brightness and yellowness levels in the LAB color space. This combination allowed for effective teeth whitening without requiring a fully deep learning-based pipeline.

The whitening function workflow includes:

1. Resizing the segmentation mask to match the dimensions of the original image.
2. Converting the original image to the LAB color space to isolate luminance and chrominance channels.
3. Enhancing brightness and reducing yellowness using arithmetic operations.
4. Blurring the mask for smooth blending and overlaying the adjusted teeth region back onto the original image.

2.3 Traditional Techniques

In the Hollywood smile component, a projection-based approach was used as an alternative to deep learning. This method aimed to align and enhance the smile by projecting a 3D teeth model onto the image. However, this approach was motivated by the absence of suitable datasets required for deep learning-based solutions.

2.4 Eye Detection and Face Alignment

A module was designed to detect the center of the eyes and determine any inclination of the face. This information would enable automatic rotation of the image to ensure horizontal alignment, optimizing the projection process.

To ensure accurate projection-based enhancements. This module was designed to:

- Detect the centres of the eyes using traditional algorithms for eye-detection.
- Calculate the inclination of the line connecting the eye centres.
- Apply a rotation transformation to align the image horizontally.

While the module showed promising results in aligning images, its integration was limited due to the constraints of other modules in the pipeline, which hindered the overall Hollywood smile enhancement process. But this module is ready to be used in future if needed.



Figure 2.1: Face is level.



Figure 2.2: Face is inclined with slope 0.26.

Figure 2.3: Outputs of pupil-detection algorithms.

Chapter 3

Results

3.1 Outcomes

The segmentation-based teeth bleaching process demonstrated exceptional visual and quantitative results, highlighting the efficacy of the applied deep learning model. Conversely, the Hollywood smile component encountered challenges due to limitations in segmentation precision. Specifically, the inability of the model to isolate individual teeth led to bad results with the projection-based enhancement approach. This limitation underscores the need for a refined segmentation framework to achieve tooth-level granularity.

3.2 Sample Outputs

3.2.1 Segmentation Module Output

The segmentation module played a pivotal role in the teeth bleaching process, generating masks to isolate regions of interest. For more details about the segmentation model, refer to Appendix A. Below are sample outputs demonstrating its ability in Figures 3.3, 3.6

3.2.2 Bleaching Results

The teeth bleaching results were achieved through a hybrid approach that combined the segmentation module with a traditional whitening function. Powered by the BiSeNet model, the segmentation module effectively isolated the teeth region. The resulting masks were subsequently processed using a custom whitening function that applied brightness and yellowness adjustments within the LAB color space.



Figure 3.1: Input image for segmentation.

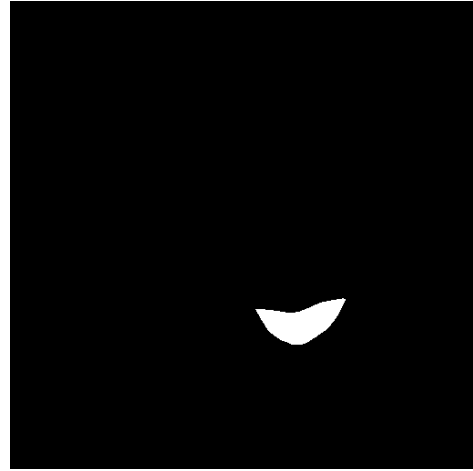


Figure 3.2: Segmentation output mask generated by the BiSeNet model.

Figure 3.3: Comparison of the input image and the corresponding segmentation output mask.



Figure 3.4: Input image for segmentation.

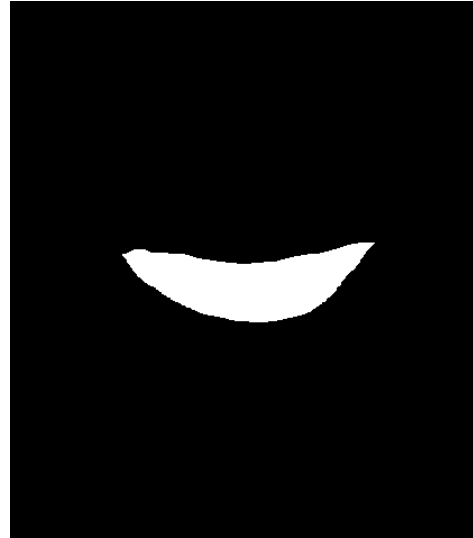


Figure 3.5: Segmentation output mask generated by the BiSeNet model.

Figure 3.6: Comparison of the input image and the corresponding segmentation output mask.

This approach produced visually satisfactory results, demonstrating the effectiveness of combining deep learning for segmentation with traditional image processing techniques. Further details about the whitening process are available in Appendix B. Results for Figure 3.1 are presented in Figure 3.7, while the outcomes corresponding to Figure 2.2 are shown in Figure 3.8.



Figure 3.7: Output 1: Final output demonstrating enhanced teeth brightness and reduced yellowness post-bleaching.



Figure 3.8: Output 2: Final output demonstrating enhanced teeth brightness and reduced yellowness post-bleaching.

Figure 3.9: Result of the teeth bleaching process.

3.2.3 Hollywood Smile Challenges

See the result of Figure 3.4 in Figure 3.12, Note the issue also with the 3D model it doesn't contain any texture as shown in 3.11.

While the projection approach attempted to address the data limitation challenge, it revealed several critical issues that hindered the success of the Hollywood smile enhancement process:

1. **Inability to Segment Individual Teeth:**

The segmentation module, designed for teeth bleaching, **lacked the granularity to isolate individual teeth**. Instead, it **treated the**



Figure 3.10: Projection output.

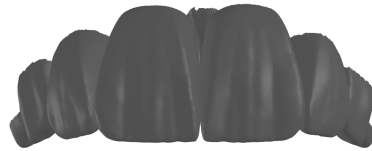


Figure 3.11: 3D template.

Figure 3.12: Projection results.

entire oral cavity as a single region, which was sufficient for bleaching but unsuitable for the Hollywood smile process. This limitation made it impossible to achieve the level of precision required for projecting a 3D model of teeth onto the image.

2. Uncertainty in Tooth Count and Identification:

Even if segmentation masks were refined, there was no mechanism to count or uniquely identify individual teeth. This lack of specificity created alignment issues when attempting to map the 3D model vertices to the corresponding regions in the image.

3. Missing Tooth-Level Vertex Mapping in the 3D Model Template:

The 3D model used for projection lacked tooth-specific vertex mappings, which are essential for aligning the model with the segmented teeth. Without precise **vertex-to-image mapping**, the projection was prone to inaccuracies, leading to bad results.

4. Data Limitations, Quality and Diversity:

The absence of a dedicated dataset for segmentation. Hollywood smile enhancements posed a significant challenge. **Without training data to fine-tune the segmentation module for tooth-level precision or improve the 3D model mappings**, the overall pipeline was

constrained. The lack of diverse and annotated datasets for training limited the segmentation model’s performance. This deficiency hindered its generalization to different dental configurations and imaging conditions.

5. Inconsistent Results from Projection-Based Techniques:

Projection techniques rely heavily on the quality of both segmentation and the 3D model template. The combined limitations of segmentation and template granularity led to inconsistent and bad outcomes in smile enhancements.

6. Dependence on Projection-Based Techniques:

The reliance on projection-based methods, in the absence of granular segmentation, introduced limitations. These methods are highly sensitive to inaccuracies in segmentation and alignment, resulting in inconsistent and often visually unacceptable outputs.

7. Projection Artifacts:

Misalignment during the projection process introduced visual artifacts such as overlapping or misplaced teeth regions, as shown in Figure 3.12. These issues further degraded the quality of the Hollywood smile output.

8. Data Limitations for Deep Learning Approaches:

The absence of a suitable dataset with annotated individual teeth restricted the exploration of alternative deep-learning models that could potentially address these challenges.

9. Lack of Validation Metrics:

The project lacked robust quantitative metrics to validate the alignment and aesthetic quality of the Hollywood smile outputs. This made it difficult to systematically identify and rectify the sources of error.

10. Integration Challenges: The segmentation inaccuracies compounded errors in subsequent modules, such as the face alignment and projection stages, reducing the overall effectiveness of the pipeline.

These issues underscore the limitations of the current approach and highlight the need for **more advanced segmentation techniques** and **refined 3D model templates** or **comprehensive datasets**, and enhanced 3D modeling for successful Hollywood smile enhancements to achieve the desired results.

Chapter 4

Discussion

4.1 Challenges

The projection-based approach faced significant challenges due to the segmentation module’s limitations and the lack of tooth-level granularity in the 3D model template. These issues led to imprecise results, reducing the effectiveness of the Hollywood smile enhancement process.

4.2 Potential of Face Alignment Module

The face alignment module offers a robust solution for improving projection-based methods. By ensuring precise alignment, it could minimize distortions and enhance the accuracy of smile enhancements. However, its effectiveness is contingent on improvements in segmentation and data quality, underscoring the need for a comprehensive pipeline.

4.3 Future Work

To address the challenges identified in the proof of concept, future efforts should focus on the following directions:

1. **Enhancing Segmentation Granularity:** Develop or fine-tune segmentation models to achieve individual tooth-level granularity. Multi-task learning or incorporating contextual cues (e.g., jaw structure) can improve segmentation precision. Additionally, semi-supervised learning approaches could be employed to make better use of limited annotated data.

2. **Connected Component Analysis for Tooth Segmentation:** Leverage connected component algorithms to identify individual teeth from a binary mask of the teeth region. The steps include:

- (a) Convert the segmentation mask into a binary image.
- (b) Apply connected component labeling to separate contiguous regions.
- (c) Use geometric features (e.g., size, shape, position) to filter and classify components as individual teeth.

This method can help isolate teeth effectively in cases where segmentation models treat the teeth as a single region.

3. **Augmenting the 3D Model Template:** Incorporate metadata into the 3D model template that maps vertices to specific teeth or dental regions. This enhancement will enable precise alignment between the segmented teeth and the 3D model.

4. **Hybrid Approaches:** Explore the integration of projection-based methods with transfer learning or advanced image processing techniques. This could include using pre-trained models on similar datasets and fine-tuning them for tooth segmentation or projection tasks.

5. **Detecting Middle Teeth Corners:** Utilize corner detection algorithms (e.g., Shi-Tomasi) to detect specific features of the middle teeth for tasks such as alignment and geometry-based enhancements. The steps include:

- (a) Restrict the detection area to the central fraction of the image to isolate the middle teeth.
- (b) Use corner detection techniques to identify key points for precise alignment.
- (c) Combine these points with segmentation outputs for improved projection accuracy.

6. **Advanced Feature Detection for Smile Alignment:** Investigate traditional and machine learning-based approaches for feature detection, such as Harris Corner Detection for identifying critical points on teeth.

7. **Geometric Models for Teeth Estimation:** Leverage geometric models to estimate the number and positions of teeth, even without detailed segmentation. This can be achieved by:

- (a) Using the detected contours to approximate the shape and size of individual teeth.
 - (b) Fitting geometric primitives (e.g., ellipses) to represent each tooth region.
8. **Enhanced Projection Methods:** Develop projection methods that utilize advanced transformation techniques, such as:
- Enhancement on Iterative Closest Point (ICP) algorithms for aligning 3D models with segmented regions.
 - Non-rigid transformations to adapt the 3D model to the variability in tooth shapes and sizes.
9. **Exploring Alternative Segmentation Models:** Evaluate newer architectures like Transformers or UNet++ for dental segmentation, potentially improving the ability to isolate individual teeth.

These proposed directions aim to address the current limitations while exploring innovative techniques for dental image processing and Hollywood smile enhancement.

Chapter 5

Conclusion

While some explored algorithms and methods did not yield the expected results, this report focuses on the most effective approaches and highlights their contributions to the project’s objectives. The challenges encountered in the Hollywood smile component, particularly in segmentation and projection accuracy, provide valuable insights for future improvements.

This proof of concept successfully demonstrated a hybrid approach that combines deep learning and traditional techniques for dental image processing. The teeth bleaching component achieved highly satisfactory results, leveraging the BiSeNet segmentation model to isolate the teeth region and applying a traditional whitening function to enhance brightness and reduce yellowness. This hybrid methodology highlighted the effectiveness of combining computationally efficient traditional algorithms with modern deep learning models.

Conversely, the Hollywood smile component faced significant challenges, underscoring the limitations of the current pipeline. The inability to segment individual teeth hindered precise alignment and projection of the 3D model. Additionally, the lack of suitable datasets and metadata for the 3D model template further restricted progress. These constraints resulted in bad outcomes, with misalignments and visual artifacts impacting the aesthetic quality of the enhancements.

To overcome these limitations, future work should prioritize:

- Enhancing segmentation granularity to isolate individual teeth.
- Incorporating tooth-specific vertex mappings into the 3D model template.
- Exploring innovative traditional and machine learning-based algorithms to improve alignment and projection accuracy.

- Addressing data scarcity through synthetic data generation and semi-supervised learning techniques.

Despite these challenges, the POC has laid a strong foundation for further advancements in dental image processing. By integrating deep learning, traditional algorithms, and novel methodologies, the project demonstrates the potential to achieve personalized and aesthetically accurate dental enhancements. With the outlined improvements, future iterations could bridge the existing gaps, paving the way for practical applications in cosmetic dentistry.

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Appendix A

Details of the Segmentation Model

The segmentation module employs a state-of-the-art architecture known as **BiSeNet (Bilateral Segmentation Network)** Yu et al. 2018 Yu et al. 2020. The primary objective of this module is to isolate regions of interest in dental images, enabling focused processing for applications like teeth bleaching and Hollywood smile enhancement. Below, we describe the model in detail:

A.1 Architecture Overview

- **Backbone:** A ResNet-18 network is utilized as the feature extractor. It captures multi-scale semantic features at different resolutions ($1/8$, $1/16$, and $1/32$ of the input size).
- **Context Path:** Incorporates attention refinement modules (ARMs) to enhance semantic features from the ResNet backbone.
- **Feature Fusion Module (FFM):** Combines spatial and semantic features to produce refined feature maps for segmentation.
- **Output Heads:** Provides predictions at multiple scales to facilitate multi-scale supervision during training.

A.2 Forward Pass Workflow

1. Input images are passed through the ResNet-18 backbone to extract features at multiple levels ($feat8$, $feat16$, $feat32$).

2. The Context Path refines features using ARMs and progressively upsamples them to merge multi-scale information.
3. Spatial features from the ResNet backbone are fused with context features using the FFM.
4. The fused features are passed through segmentation heads to produce output predictions at different resolutions.

A.3 Challenges and Limitations

- The model segments the entire oral cavity rather than isolating individual teeth, which impacted the Hollywood smile enhancement process.
- Future improvements should focus on enhancing the segmentation granularity to achieve tooth-level precision.

A.4 Architecture

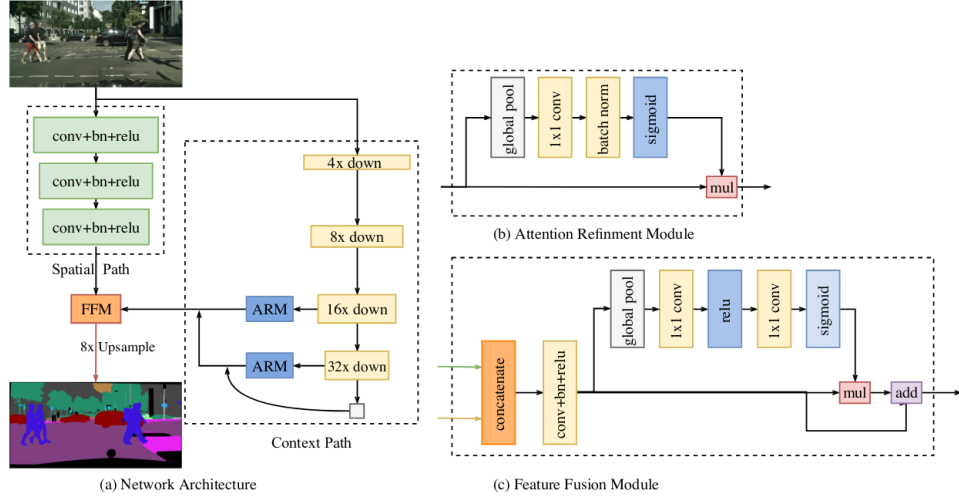


Figure A.1: Architecture of the BiSeNet model used for segmentation.

Appendix B

Traditional Techniques Details

This appendix outlines the traditional algorithms used in the proof of concept for tasks such as teeth whitening and eye detection. These algorithms were implemented alongside deep learning techniques to address specific challenges in the project.

B.1 Teeth Whitening Algorithm

The teeth whitening process involved adjusting the brightness and reducing the yellowness of the teeth region in the LAB color space. The steps are as follows:

1. Resize the segmentation mask to match the dimensions of the original image.
2. Convert the image to the LAB color space.
3. Split the LAB channels to isolate the luminance (L) and chrominance (b) components.
4. Increase brightness by adding a constant value to the L channel.
5. Reduce yellowness by subtracting a constant value from the b channel.
6. Merge the modified LAB channels and convert back to the RGB color space.
7. Smooth the segmentation mask using Gaussian blur and blend the adjusted region with the original image.

This algorithm enhanced the visual appeal of teeth without relying on computationally intensive methods.

B.2 Eye Detection and Face Alignment

A pupil detection algorithm was employed to identify the center of each eye for face alignment. The process involves:

1. Convert the eye region of the image to grayscale.
2. Apply Gaussian blur to reduce noise and improve contour detection.
3. Use a binary threshold to detect darker regions corresponding to the pupil.
4. Find contours and select the largest one, assuming it represents the pupil.
5. Calculate the center of the selected contour using image moments.
6. Align the face horizontally by rotating the image based on the line connecting the detected pupil centers.

This method provided a computationally efficient solution for improving face alignment.

B.3 Integration of Traditional Algorithms

These traditional algorithms were combined with deep learning modules to achieve the desired outcomes:

- The teeth whitening algorithm used the segmentation mask generated by the BiSeNet model.
- The face alignment module assisted in enhancing the accuracy of projection-based methods in the Hollywood Smile component.

By leveraging both traditional and deep learning approaches, the proof of concept demonstrated a hybrid methodology that balances computational efficiency and performance.